



**DAFFODIL INTERNATIONAL UNIVERSITY**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**  
**FYDP (TITLE PHASE) EVALUATION REPORT**

**REPORTING PERIOD- SUMMER 2025**

**Project Identification:**

|                            |                                                                                                                                                                                                                                                                                          |                                             |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>I. Project Title</b>    | <b>“A Hybrid Explainable AI Framework for Automatic Tuberculosis Detection from Chest X-Ray Images Using Convolutional Neural Networks”</b>                                                                                                                                              |                                             |
| <b>II. Group Members</b>   | <b>ID: 0242310005101019</b>                                                                                                                                                                                                                                                              | <b>Name: Shariar Ahamed Ripon</b>           |
|                            | <b>ID: 0242310005101020</b>                                                                                                                                                                                                                                                              | <b>Name: Md.Moniruzzaman Rifat</b>          |
| <b>III. Supervisor</b>     | <b>Name: Dr. Md. Ali Hossain</b>                                                                                                                                                                                                                                                         | <b>Designation: Associate Professor</b>     |
| <b>IV. Co-Supervisor</b>   | <b>Name: Mr. Md. Mizanur Rahman</b>                                                                                                                                                                                                                                                      | <b>Designation: Lecturer (Senior Scale)</b> |
| <b>V. Submission Date:</b> | <b>22-04-2026</b>                                                                                                                                                                                                                                                                        |                                             |
| <b>VII. Certificate:</b>   | <b>“This is to certify that the final year design project work until Title defense evaluation held on _____, titled as stated in Sec. I, executed by the students group mentioned in Sec. II, have been found satisfactory and every section of this report is reflecting the same.”</b> | <b>(Signature of Supervisor &amp; date)</b> |

## CO Description and Corresponding PO for Title Defense Phase

| CO  | CO Descriptions                                                                                                                                                                                                                                                                | PO  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| CO1 | Integrate recently gained and previously acquired knowledge to identify a real-life complex engineering problem for the “ <b>A Hybrid Explainable AI Framework for Automatic Tuberculosis Detection from Chest X-Ray Images Using Convolutional Neural Networks</b> ” project. | PO1 |
| CO2 | Analyze different aspects of the goals in designing an engineering problem solution for the “ <b>A Hybrid Explainable AI Framework for Automatic Tuberculosis Detection from Chest X-Ray Images Using Convolutional Neural Networks</b> ” project.                             | PO2 |
| CO3 | Explore diverse problem domains through a literature review, delineate the issues, and establish the goals for the FYDP.                                                                                                                                                       | PO4 |

### 1. Introduction:

Tuberculosis (TB) is a serious infectious disease that remains a major global health concern, especially in developing countries. Early and accurate detection of TB is essential to reduce its spread and improve patient outcomes. Chest X-ray imaging is one of the most commonly used diagnostic tools for TB screening due to its low cost and wide availability. However, manual interpretation of X-ray images by radiologists can be time-consuming and may lead to errors, especially in high patient-load environments. [1]

Recent advancements in Artificial Intelligence, particularly deep learning, have shown promising results in automating medical image analysis. Among these techniques, Convolutional Neural Networks (CNNs) are widely used for image classification tasks, including disease detection. However, many existing AI-based systems lack transparency, making it difficult for medical professionals to trust their decisions. [2]

This project proposes a hybrid explainable AI framework that uses CNNs along with attention-based visualization techniques to automatically detect TB from chest X-ray images while providing interpretable results. The system also aims to compare its performance with radiologist-level benchmarks.

### 2. Background Study:

In recent years, the application of Artificial Intelligence in healthcare has significantly increased, particularly in medical image analysis. One of the key techniques used in this domain is Deep Learning, which allows models to automatically learn patterns from large datasets. Within deep

learning, Convolutional Neural Networks (CNNs) have become the most popular method for analyzing image data, including chest X-rays. [3]

Several studies have demonstrated that CNN-based models can achieve high accuracy in detecting tuberculosis from chest X-ray images. These models are trained on large datasets and are capable of identifying subtle patterns that may not be easily visible to human observers. As a result, automated TB detection systems have the potential to assist radiologists and improve diagnostic efficiency. [4]

However, despite their high performance, most CNN-based models operate as black-box systems. This means that they provide predictions without explaining how the decision was made. In clinical environments, this lack of interpretability is a major limitation, as healthcare professionals need to understand the reasoning behind a diagnosis before trusting the system. [5]

To address this issue, researchers have introduced explainable AI techniques such as attention maps and Grad-CAM. These methods highlight the regions of an image that contribute most to the model's prediction, making it easier for doctors to interpret the results. Although these techniques have been applied in various medical imaging tasks, their integration into TB detection systems is still limited. [6]

Furthermore, many existing studies focus mainly on improving model accuracy but do not perform proper benchmarking against radiologist performance. Without such comparison, it is difficult to evaluate the practical usefulness of these systems in real-world clinical settings.

Therefore, there is a need for a system that not only provides accurate TB detection but also ensures transparency and clinical relevance. This project aims to combine CNN-based detection with attention-based explainability and benchmarking to address these limitations.

### **3. Problem Statement:**

Tuberculosis detection using chest X-ray images relies heavily on manual interpretation by radiologists, which can be time-consuming and prone to human error. Although AI-based models have been developed to automate this process, many of them function as black-box systems without providing clear explanations for their predictions. This lack of interpretability reduces trust and limits their use in clinical environments.

Additionally, existing systems often focus only on accuracy and do not compare their performance with radiologists, making it difficult to evaluate their real-world effectiveness. Therefore, there is a need for an explainable and reliable AI-based system that can provide accurate TB detection along with interpretable results and proper benchmarking.

#### **4. Preliminary Novelty Statement:**

This project is unique because it combines accurate TB detection with explainable AI techniques in a single framework. Unlike traditional models, the proposed system will provide visual explanations using attention maps, helping users understand the decision-making process. Additionally, the inclusion of benchmarking against radiologist performance adds practical value and improves the reliability of the system.

#### **5. Research Questions:**

Does a hybrid CNN-based explainable AI framework improve tuberculosis detection accuracy and interpretability from chest X-ray images compared to conventional automated methods?

#### **6. Objectives:**

- To study existing AI-based methods for tuberculosis detection
- To design a CNN-based framework for automatic TB detection
- To integrate attention-based explainability techniques
- To evaluate the model using standard performance metrics
- To compare the system performance with radiologist benchmarks

#### **7. Expected Outcome:**

The expected outcome of this project is an efficient and explainable AI-based system for tuberculosis detection. The system will provide accurate classification results along with visual explanations, improving transparency and trust. It will also demonstrate competitive performance when compared to radiologist-level results, making it useful for real-world healthcare applications.

#### **8. Scope of Work:**

This project will focus on designing and evaluating an AI-based framework for TB detection using chest X-ray images. The study will use publicly available datasets and standard evaluation metrics. It will not involve real-time clinical deployment or patient data collection. The project is limited to model design, explainability, and performance comparison within a research environment.

## 9.Reference

- [1] G. Siddharth, "Enhanced CoAtNet based hybrid deep learning architecture for automated tuberculosis detection in human chest X-rays," *BMC Medical Imaging*, vol. 25, no. Springer, p. 379, 2025.
- [2] Y. H. Bhosale, "Thoracic-net: Explainable artificial intelligence (XAI) based few shots learning feature fusion technique for multi-classifying thoracic diseases using medical imaging," *Multimedia Tools and Applications*, vol. 84, no. Springer, pp. 5397-5433, 2025.
- [3] K. Vanitha, "Enhanced tuberculosis detection using Vision Transformers and explainable AI with a Grad-CAM approach on chest X-rays," *BMC Medical Imaging*, vol. 25, no. Springer, p. 96, 2025.
- [4] B. Nansamba, "A systematic review on application of multimodal learning and explainable AI in tuberculosis detection," *IEEE Access*, no. IEEE, 2025.
- [5] S. I. Nafisah, "Tuberculosis detection in chest radiograph using convolutional neural network architecture and explainable artificial intelligence," *Neural Computing and Applications*, vol. 36, no. Springer, pp. 111-131, 2024.
- [6] O. Sarkar, "Multi-scale CNN: An explainable AI-integrated unique deep learning framework for lung-affected disease classification," *Technologies*, vol. 11, no. MDPI, p. 134, 2023.

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| <b>Supervisor's Feedback:</b>                                                                                                             |
| Provide comments and suggestions for improvement                                                                                          |
| <div style="margin-top: 10px;">1. _____</div> <div style="margin-top: 10px;">2. _____</div> <div style="margin-top: 10px;">3. _____</div> |

**Examiner's Signature:** \_\_\_\_\_

**Date:** \_\_\_\_\_